

In-Class Template

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1. Set Up

Packages

```
# general use and cleaning

library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))
```

```

#survey data (reading different sheets in as seperate data frames)
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  # values in df that are actually indicating missing values, making it seem as a chr when :
  # different ways missing values show up in data set
  na = c("over", "", "n/a", "N/A", "173+")
)

```

2. Cleaning

sites object

Goal: create a new object from **sites** that only includes NCOS sites

```

# creating new object from sites
ncos_sites <- sites |>
  # filtering to only include NCOS quarterly sites
  filter(sector == "NCOS" &
    sampling_frequency == "Quarterly")

```

clean taxn_list

Goal: convert chr strings in the existing **taxon_list** df into snakecase

```

# creating new object from taxon_list
taxon_list_clean <- taxon_list |>
  # converted taxon names to snake case

```

```
mutate(verbatim_name = to_any_case(verbatim_name,
                                   case = "snake"))
```

clean water_quality

Goal: to create a df with median pH, median DO (mg/L), median salinity (ppt) for each data and site

- create new object from `water_quality`
- clean column names
- only include surveys from quarterly NCOS sites
- create a new column called `date_site` to join with later data frames
- group_by date_site
- calculate median pH, salinity, and DO -ungroup data frame -filter out any outliers

```
# creatign new object from water_quality
water_quality_clean <- water_quality |>
# cleaning column name
clean_names() |>
# a filtering join, only keep the rows in wq clean in which the site matches up with the s
semi_join(ncos_sites, by = "site") |>
# creating new column to join with later data frames
unite("date_site", date, site, remove = TRUE) |>
# grouping by date/site
group_by(date_site) |>
# calculating the medians
summarize(med_pH = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungrouping df
ungroup() |>
# filtering out outliers
filter(date_site != "2022-08-12_NVBR")
```

clean aq_ins

Goal: create a long format data frame (currently in wide format, like lots of other biological data sets) in which each row contains the counted individuals in each taxon during each survey. The data frame should include taxonomic information for each organism and the water quality information for each survey.

new df should have more rows

functions in R expect long format

- create a new object from `aq_ins`
- clean column names
- filter tp only include quarterly NCOS sites
- select columns
- filter to only include “filtered beaker” sample
- convert df to long format
- group by ste date and taxon
- calculate avg abundance per liter (sum counts divide dby 7.5 liters)
- create a new column called `date_site` to join with water quality
- join data frames
- join with taxon lists

```
# create a new object
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join only include quarterly ncoss surveys
semi_join(ncos_sites, by = "site") |>
# renaming date_on_vial column to date
rename(date = date_on_vial) |>
# selecting columns of interest
# keeping information about each survey and the 9 most abundant taxa
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filtered to only include filtered beaker samples (better sampling technique)
filter(sample_type %in% c("FB 250", "FB250")) |>
# converting to long format, making df longer
pivot_longer(
  # name the columns to make longer
  cols = ostracod:annelida_polychaete,
  # new column for old column names
```

```

# column names are all in new column called taxon
names_to = "taxon",
# new column for the values in each of the old columns
# counts now under a column called abundance
values_to = "abundance"
) |>
# group by date site and taxon
group_by(date, site, taxon) |>
# calculated average count per liter (total abundance/7.7)
summarise(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungrouping data frame
ungroup() |>
# new column to join data frames, didnt remove them
unite("date_site", date, site, remove = FALSE) |>
# joined with water quality clean
left_join(water_quality_clean, by = "date_site") |>
# joined with taxonomic information
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# fill in full site names
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "East Channel",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# set factor levels for full site name by median salinity
mutate(site_full = fct_reorder(site_full,
                               med_sal,
                               .fun = "median",
                               .na_rm = TRUE),
       # reversing the order
       site_full = fct_rev(site_full))

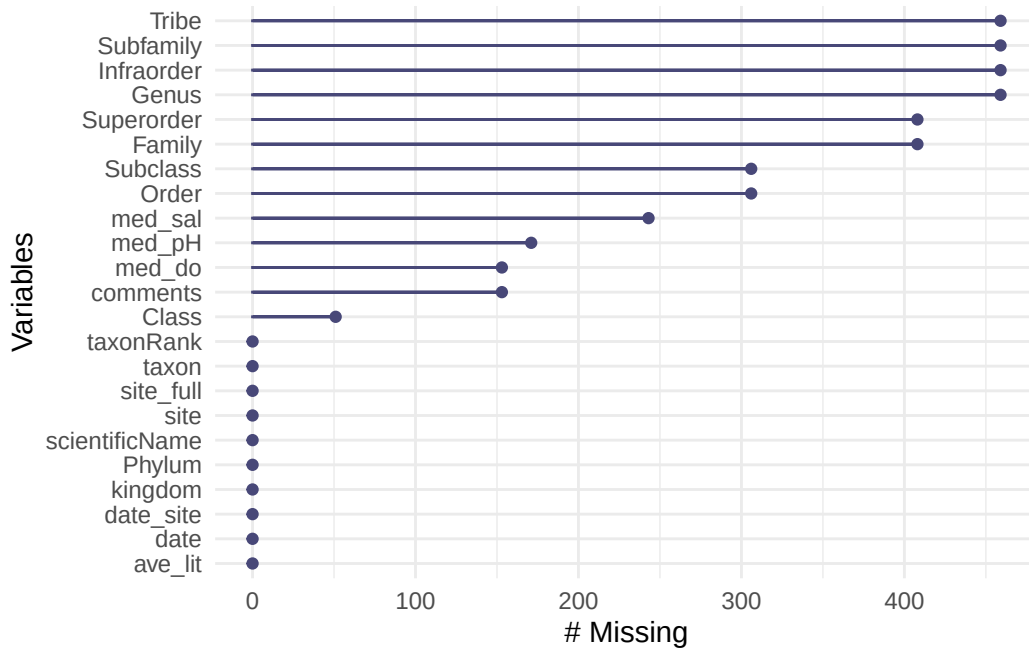
# wide to long, making missing values explicit

```

3. Visualizing

Visualize Missing Values

```
gg_miss_var(aq_ins_clean)
```



Missing values for salinity, pH, and DO might influence later figures

Visualize Salinity

- x: axis: site
- y:axis: mean salinity
- geometry to show salinity during survey: `geom_jitter()` - geometry to show median salinity: `stat_summary`

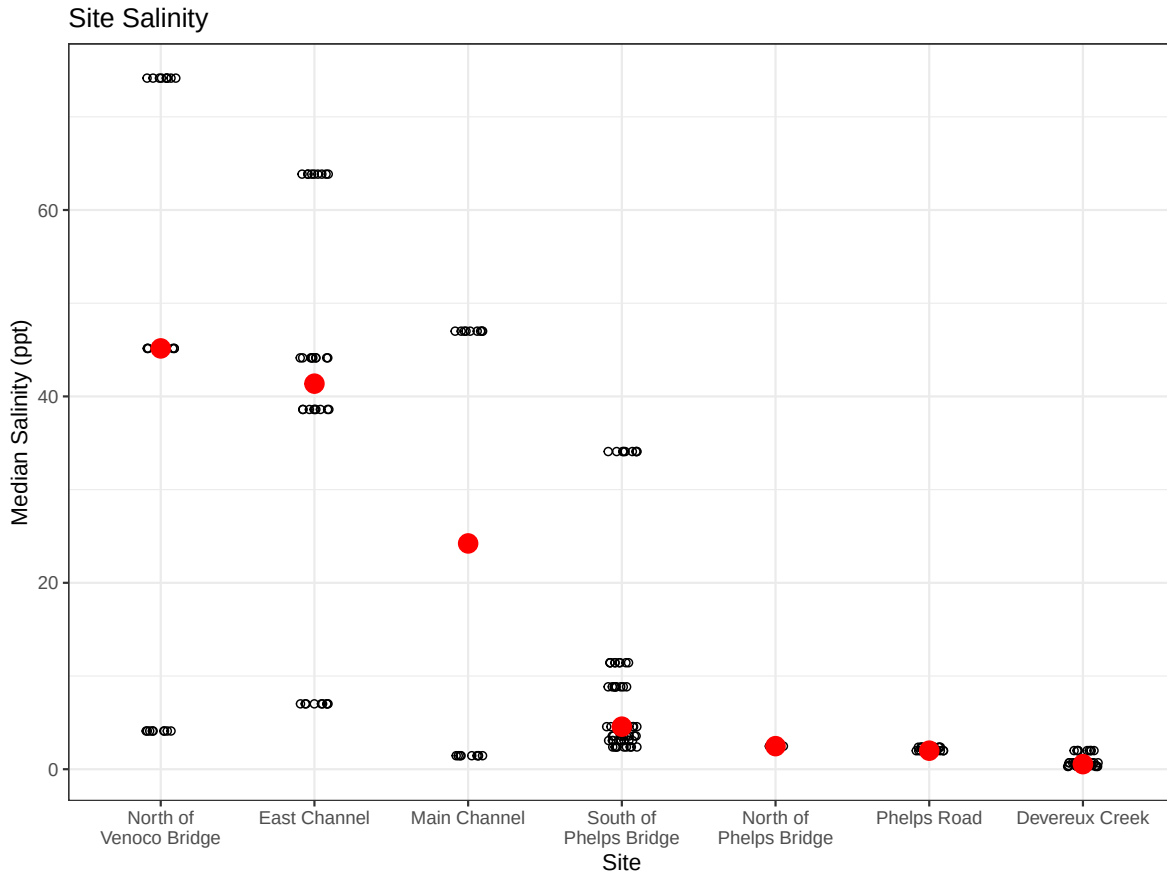
```
# creating new object to store plot
salinity_plot <- ggplot(data = aq_ins_clean,
                        mapping = aes(x = site_full,
                                      y = med_sal)) +
  # points representing median salinity at each survey
  geom_jitter(height = 0,
```

```

        width = 0.1,
        shape = 21) +
# points representing mean salinity
stat_summary(geom = "point",
             fun = median,
             color = "red",
             size = 4) +
# adding line breaks to x axis labels (wrapping)
scale_x_discrete(labels = label_wrap(width = 15)) +
# relabeling axes and adding a title
labs(x = "Site",
     y = "Median Salinity (ppt)",
     title = "Site Salinity") +
# adding themeing
theme_bw()

# visualizing the object
salinity_plot

```



Sites seem to differ in salinity. North of Venoco bridge, East Channel, and Main Channel tend to be hyper-saline while North of Phelps Bridge, Phelps Road, and Devereux Creek is the least saline. South of Phelps bridge tends to be more fresh water but it higher in salinity than the other, lower salinity sites

Visualize differences in copepod abundance

- x-axis: site_full
- y-axis: average_abundance_l
- geometry to represent observations: geom_jitter
- geometry to represent medians: stat_summary

Initial Cleaning

```

copepods <- aq_ins_clean |>
  filter(taxon == "copepod") |>
  # adding 1 bc some observations are 0
  # helps overcome issues with large differences in order of magnitude
  mutate(log_ave_lit = log(ave_lit + 1))

```

Visualizing

```

# creating object to store plot
copepod_plot <- ggplot(data = copepods,
                      mapping = aes(x = site_full,
                                    # using log transformed count per liter
                                    y = log_ave_lit)) +

  # jitter to represent each sample
  geom_jitter(height = 0,
             width = 0.1,
             shape = 21) +

  # points to represent median copepod abundance
  stat_summary(geom = "point",
             fun = median,
             color = "blue",
             size = 2) +

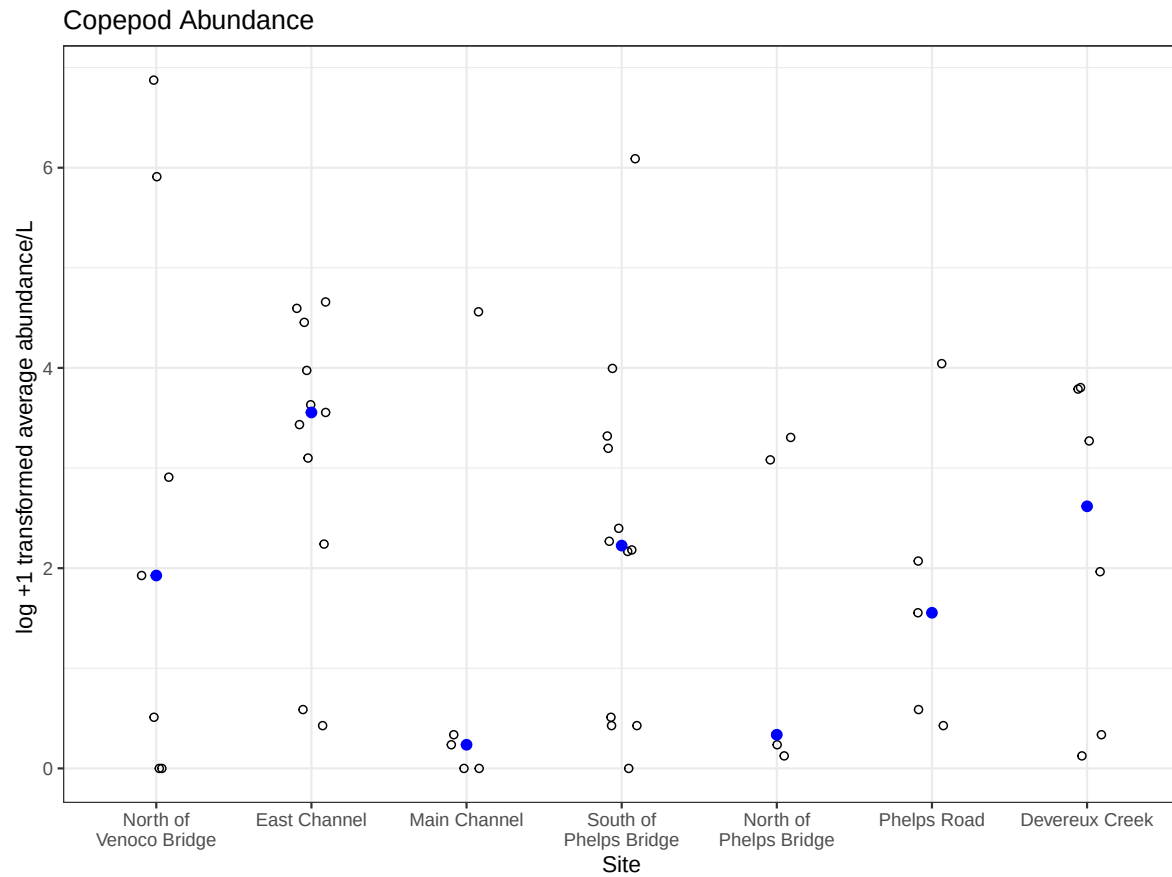
  # wrap x-axis labels
  scale_x_discrete(labels = label_wrap(width = 15)) +

  # relabeling axes
  labs(x = "Site",
       y = "log +1 transformed average abundance/L ",
       title = "Copepod Abundance") +

  # using different theme
  theme_bw()

# displaying the object
copepod_plot

```



Copepod abundance differs across sites. Main Channel, North of Phelps tend to be low N of Venoco bridge east channel,...

Putting Salinity and Copepod abundance together

```
salinity_plot / copepod_plot
```

